**Sheet 22** Thu. 10 Dec. 2026 Square root and cube; sign of the reference functions

**1.** Warm-up · Flash Calculate $\overline{49}$, $\sqrt{0.25}$ and $(-2)^3$.

**2.** Core · Method Study the sign of $f(x) = -2x + 6$.

**3.** Core + · True/false True or false? Justify. For every real number $x$, $\sqrt{x^2} = x$.

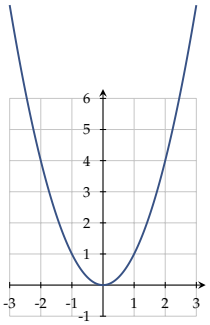
**4.** Review · Flash How many squares make up a net of a cube? Does a cross (row of 4 squares + 1 square on each side) work?

**5.** Running thread · Flash How many faces, vertices and edges does a square-based pyramid have?

**Sheet 23** Mon. 14 Dec. 2026 Graphical and numerical solving: f $(x) = k$, $f(x) < k$

**1.** Warm-up · Flash Solve $x^2 = 16$ then $x^2 = -4$.

**2.** Core · Graph Graph of $f(x) = x^2$ :




Solve graphically $f(x) = 4$ then $f(x) < 4$.

**3.** Core + · Method Solve $|x| < 3$ then $|x| = 5$.

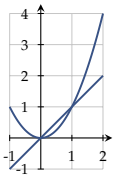
**4.** Review · Pitfall Calculate $-x^2$ then $(-x)^2$ for $x = -3$.

**5.** Running thread · Method In a triangle $ABC$, $M$ & $N$ are the midpoints of $[AB]$ & $[AC]$. What is being proved? Which property is being cited?

**Sheet 24** Tue. 15 Dec. 2026 Solving f $(x) = g(x)$ *et* $f(x) < g(x)$

**1.** Warm-up · Flash Solve 2 $x - 1 = x + 3$.

**2.** Core · Graph Graphs of $f(x) = x$ and $g(x) = x^2$ :




Solve $f(x) = g(x)$ then $f(x) < g(x)$.

**3.** Core + · Reading code Output of `f(-1) == g(-1)` ? And `f(0) == g(0)` ?

```
def f(x):
    return x*x
def g(x):
    return 2*x + 3
```

**4.** Review · Problem Plan A: 15,000 Ar fixed + 1,000 Ar per session. Plan B: 2,500 Ar per session. From how many sessions onwards is A cheaper?

**5.** Running thread · Flash Calculate the coordinates of the midpoint of $[AB]$ with $A(-2;5)$ and $B(6;1)$.

## One-variable descriptive statistics

**Sheet 25** Thu. 17 Dec. 2026 Mean and linearity of the mean

**1.** Warm-up · Flash Mean of 12, 15, 9, 14?

**2.** Core · Method The data set 4, 6, 8, 10, 12 has mean 8. What is the mean if we add 3 to each value? if each value is multiplied by 2?

**3.** Core + · True/false True or false? Justify. The mean of two numbers $a < b$ is always strictly between $a$ and $b$.

**4.** Review · Reading In a pie chart of 80 people, one sector represents 25%. What is its count? What is its angle?

**5.** Running thread · Flash Calculate GCD $(84\,;\,126)$ and deduce the irreducible form of $\dfrac{84}{126}$.

**Sheet 26** Mon. 4 Jan. 2027 Median and quartiles

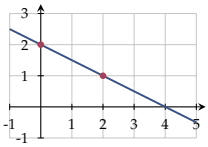
**1.** Warm-up · Flash Order 7, 2, 9, 4, 5, 11, 3 then give the median.

**2.** Core · Method Series 3; 5; 6; 8; 9; 12; 14; 15. Give $Q_1$, the median, $Q_3$ and the interquartile range.

**3.** Core + · True/false True or false? Justify. At least half of the values of a series are greater than or equal to its median.

**4.** Review · Flash Order 0.6, $\dfrac{5}{8}$ and 61% in increasing order.

**5.** Running thread · Graph → formula The line below represents an affine function $f$. Determine $f(x)$.




**Sheet 27** Tue. 5 Jan. 2027 Standard deviation; effect of adding or removing a value

**1.** Warm-up · Flash Calculate $(3-5)^2+(7-5)^2$.

**2.** Core · Method Series 2; 4; 6: mean, variance and standard deviation (approximate value to $10^{-2}$).

**3.** Core + · True/false True or false? Justify. Adding 5 to each value of a series does not change its standard deviation.

**4.** Review · Construction steps Let $ABC$ a triangle, $I$ the midpoint of $[AB]$ and $D$ the image of $C$ in in $I$. What kind of shape is $ACBD$?

**5.** Running thread · Flash $(MN) \parallel (BC)$, $M \in [AB]$, $N \in [AC]$, $AM = 3$, $AB = 5$, $AC = 10$. Calculate $AN$.

**Sheet 28** Thu. 7 Jan. 2027 Comparison of series

**1.** Warm-up · Flash Range of 3, 8, 5, 12?

**2.** Core · Interpret Two series $A$ and $B$ have the same mean 12; $\sigma_A = 1.5$ and $\sigma_B = 4$. Which is the more steadier?

**3.** Core + · Interpret $A$ : median 11, interquartile range 2. $B$ : median 13, interquartile range 6. Compare the two series.

**4.** Review · Flash Two fair coins are tossed. Probability of getting two heads?

**5.** Running thread · Method Procedure: choose a number; square it; add 1. Result for $-3$? Can we get 0? Justify.